

Enhancing Speaking and Slide Design Skills with Deep Learning: An Online Presentation Assessment System

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ABSTRACT

Presentation skills, which involve the effective use of verbal and nonverbal cues, enable audiences to better understand the content being presented. We develop a deep learning-based online assessment system that can objectively evaluate speakers' oral presentations and slide design, providing comprehensive feedback to support their self-practice. For the speaking skill assessment, we construct a multimodal neural network, including LSTMs and attention networks, to analyze the linguistic and acoustic features of oral presentations. The proposed model can predict 14 distinct types of audience impressions with an average accuracy of 85.0%. For the slide design assessment, we propose a method that can analyze slide design based on their visual and structural features, independent of file formats. It can determine whether the slides meet 10 assessment criteria with an average accuracy of 81.7%.

CCS CONCEPTS

• Applied computing → Computer-managed instruction.

KEYWORDS

Multimodal Neural Network; Online Support System; Oral Presentation; Presentation Slide

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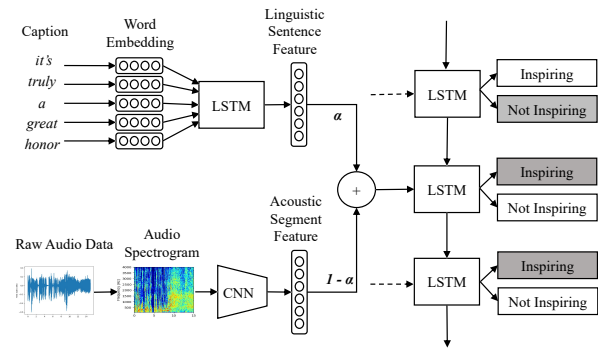


Figure 1: Speech-Language Multimodal Model.

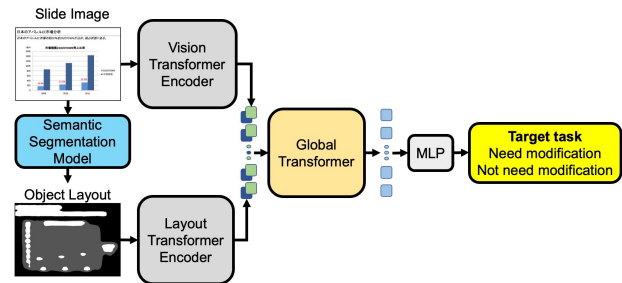


Figure 2: Vision-Layout Hierarchical Transformer.

1 INTRODUCTION

Learning presentation skills is challenging because improvement typically relies on real-life experiences and feedback. To address this issue, we propose an online presentation assessment system that supports self-practice by comprehensively evaluating speaking and slide design skills and providing criterion-based feedback.

Existing work on speaking skill assessment [9] used Support Vector Machine (SVM) and Markov Random Field (MRF) to consider the linguistic and acoustic features for predicting audience impressions. However, their model could not be fine-tuned during feature fusion. Existing work on slide design assessment [10] used the visual and structural features to evaluate the quality of slides. However, their method required PowerPoint files to extract structural features.

Task	Modality	Previous work	Ours
Speaking	Linguistic only	81.4	81.8
	Acoustic only	69.1	70.0
	Linguistic + Acoustic	83.1	85.0
Slide Design	Visual only	73.7	77.7
	Structural only	70.4	71.3
	Visual + Structural	76.0	79.1
	Visual + Structural + Imbalanced + Multi-task	81.8	81.7

Table 1: Average prediction accuracy (%) of the speaking and slide design skill assessment using different modalities.

2 SPEAKING SKILL ASSESSMENT

To address the architecture limitation of the SVM-MRF model [9], which cannot be fine-tuned during feature fusion, we proposed [11] a new Speech-Language Multimodal Model (SLMM) to predict audience impressions based on the linguistic and acoustic features. It provides presenters with an overall assessment and continuous feedback for each sentence (Figure 1).

We used the skip-gram model [7] for word embeddings and input these embeddings for each sentence into a Long Short-Term Memory (LSTM) network [6] to obtain sentence-level linguistic features. We then used a Convolutional Neural Network (CNN) to extract acoustic features from the audio spectrograms. After obtaining the sentence-level linguistic and acoustic features, we applied an attention mechanism for feature fusion. We then used an LSTM to capture the correlations between sentences and predict audience impressions for each sentence.

3 SLIDE DESIGN SKILL ASSESSMENT

Semantic Segmentation of Presentation Slides. We trained a model to perform semantic segmentation of slide objects from images without relying on PowerPoint files. We used the SPaSe dataset [5], which includes 2,000 slides from Slideshare-1M [1] with pixel-wise annotations for semantic segmentation. We examined four segmentation models in this study: PSPNet [14], OCRNet [13], BiSeNetV2 [12], and Segmenter [8]. Among the examined models, OCRNet [13] achieved the best performance with a mean Intersection over Union (mIoU) of 0.653 and a mean accuracy of 84.1%.

Vision-Layout Hierarchical Transformer. Figure 2 presents the architecture of the proposed Vision-Layout Hierarchical Transformer (VLHT) for assessing slide design. It includes two unimodal Transformer encoders [4] to extract the visual features from the slide image and the structural features from the object layout, respectively. We applied several class-imbalanced learning methods in our experiments, including class-balanced loss [3] and deferred re-weighting strategy [2]. The auxiliary task for multi-task learning is to recognize whether a slide has been revised by novices.

4 EXPERIMENTS

Speaking Skill Assessment. We collected the captions and audio data from 2,445 TED Talks presentations¹. Each presentation

¹<https://www.ted.com/talks>

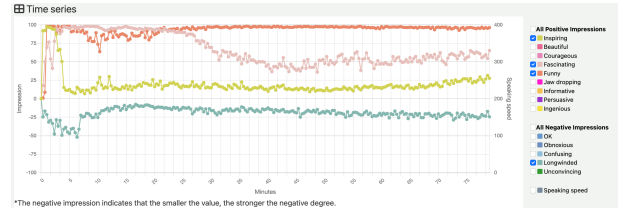


Figure 3: Example analysis of the impression prediction.

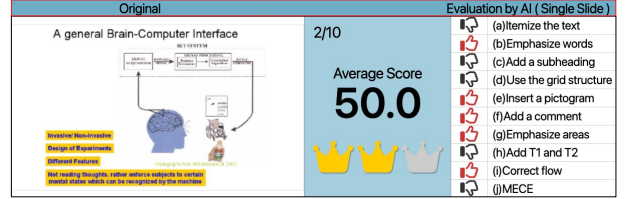


Figure 4: Example analysis of the slide design assessment.

is tagged with 14 impressions, such as *Beautiful*, *Confusing*, and *Courageous*. These tags reflect audience impressions based on their votes. Table 1 shows that the proposed method using linguistic and acoustic features achieved an average accuracy of 85.0% in predicting overall impressions. This is higher than the 83.1% average accuracy of SVM-MRF [9]. Additionally, the SVM-MRF model cannot provide continuous feedback.

Slide Design Skill Assessment. We collected 1,080 slide pairs from a training course in which novices created single-page slides and then improved them based on consultants' advice. The consultants summarized the top-10 important checkpoints in slide design, such as inserting a pictogram, adding a subheading, and emphasizing words. As shown in Table 1, when using only the structural features, the proposed method using semantic segmentation achieved an average accuracy of 71.3%, outperforming the previous work, which used one-hot vectors of sampling points to represent slide layout [10] and achieved an average accuracy of 70.4%. Furthermore, by incorporating both the visual and structural features, we applied the class-imbalanced and multi-task learning methods to achieve a final average accuracy of 81.7%.

5 DEMO SYSTEM

Our system can conduct both the overall and continuous predictions of audience impressions for the uploaded oral presentations. Figure 3 illustrates the continuous prediction results that reflect changes in audience impressions for each sentence. Users can select the impressions they are interested in to display and hide others.

Figure 4 illustrates how our system helps users identify which design problems need to be addressed. The evaluation results shows whether the slides meet the requirements of these checkpoints. The final score represents the average evaluation score across all checkpoints. Additionally, the system can highlight areas in the slides that need improvement and compare the initially uploaded slides with the latest version revised by the user.

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